

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, all set against a light green background with a subtle globe pattern.

A COMPREHENSIVE COVERAGE

MODULE - 6

BASIC CONCEPTS - 6

RELATIONSHIP BETWEEN THE SPECIES - INTERSPECIFIC INTERACTION

1 AMENSALISM

- One species harms or restricts the other species without itself being adversely affected or harmed by the presence of the other species.
- For example, the bread mould fungi *Penicillium* (Penicillium) produce penicillin, an antibiotic substance which inhibits the growth of a variety of bacteria.
- A large tree shades a small plant, retarding the growth of the small plant. The small plant has no effect on the large tree.



RELATIONSHIP BETWEEN THE SPECIES - INTERSPECIFIC INTERACTION

2 COMMENSALISM

- In this relationship one of the species benefits while the other is neither harmed nor benefited.
- Some species obtain the benefit of shelter or transport from another species. For example, sucker fish, remora often attaches to a shark by means of its sucker which is present on the top side of its head. This helps the remora get protection, a free ride as well as meal from the left over of the shark's meal. The shark does not however get any benefit nor is it adversely affected by this association.
- Another example of commensalisms is the relationship between trees and epiphytic plants.
- Epiphytes live on the surface of other larger plants and trees and use their surface for support and for obtaining sunlight and moisture. The tree gets no benefit from this relationship nor is it harmed.

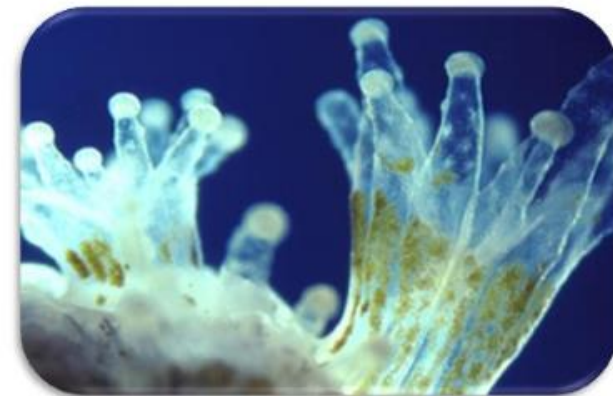


RELATIONSHIP BETWEEN THE SPECIES - INTERSPECIFIC INTERACTION

3 MUTUALISM

- This is a close association between two species in which both the species benefit.
- One example of a mutualistic relationship is that of the Oxpecker (a kind of bird) and the Rhinoceros or Zebra. Oxpeckers land on rhinos or zebras and eat ticks and other parasites that live on their skin. The oxpeckers get food and these animals get free pest control.
- However, some mutualisms are so intimate that the interacting species can no longer live without each other as they depend totally on each other to survive. Such close associations are called Symbiosis (symbiosis is intense mutualism –

Example: Coral and Zooxanthellae.

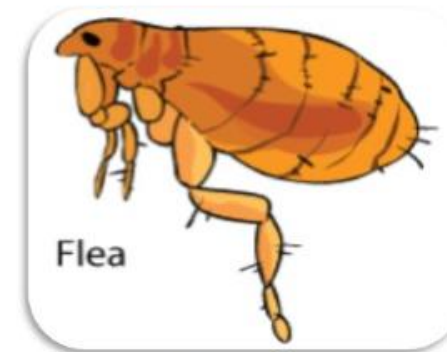


RELATIONSHIP BETWEEN THE SPECIES - INTERSPECIFIC INTERACTION

4

PARASITISM

- In this type of interaction, one species is harmed and the other benefits.
- Parasitism involves parasite usually a small organism living in or on another organism called the host from which the parasite gets its nourishment and often shelter.
- Many organisms like animal, bacteria and viruses are parasites of plants and animals.



RELATIONSHIP BETWEEN THE SPECIES - INTERSPECIFIC INTERACTION

5

PREDATION

- Predators like leopards, tigers and cheetahs use speed, teeth and claws to hunt and kill their prey.
- They keep prey populations under control. Without predators, prey species could achieve very high populations and cause ecosystem instability.



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